

UNIT I — Lecture Module 4

EVOLUTION OF CLOUD COMPUTING

INTRODUCTION:

Cloud computing is not an invention that appeared suddenly — it is the cumulative product of nearly seven decades of computing evolution. Each generation of computing technology introduced new ideas, solved the problems of the previous generation, and left behind a set of concepts that the next generation would build upon. Understanding this evolutionary journey is essential for appreciating why cloud computing is designed the way it is and why it has become the dominant paradigm for modern IT infrastructure.

The story begins in the 1950s with centralised mainframe computers shared among multiple users through time-sharing, progresses through the decentralisation of the personal computer revolution and client-server computing, accelerates through the global connectivity of the Internet, matures through the resource-pooling models of grid and cluster computing, and culminates in the virtualised, metered, on-demand service model we recognise today as cloud computing.

This lecture module traces that journey chronologically, examining the key technologies, organisations, individuals, and events that contributed to the birth and growth of cloud computing as defined by the NIST framework.

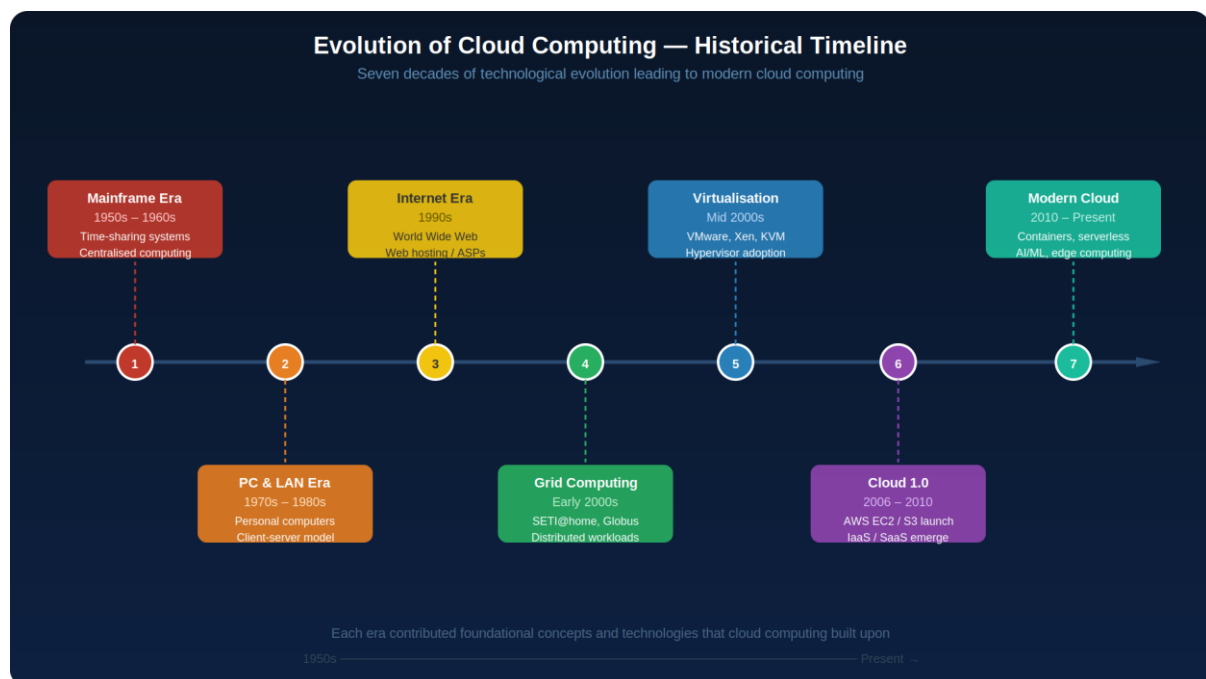


Figure 1: Historical Timeline of Cloud Computing Evolution — from mainframes (1950s) to AI-native cloud (present)

Figure 1 presents a complete timeline of the seven major eras in computing evolution. Each era is colour-coded to highlight its distinct character — from the centralised red of the mainframe era to the dynamic teal of the modern cloud. The timeline makes clear that cloud computing is not a single invention but a synthesis of ideas refined across decades.

ERA 1 — THE MAINFRAME AND TIME-SHARING ERA (1950S–1960S):

MAINFRAME COMPUTING:

The first commercial computers — mainframes — appeared in the early 1950s. These were enormous machines housed in dedicated, climate-controlled rooms and operated by specialist technicians. Early mainframes include the IBM 701 (1952) and the UNIVAC I (1951). They were extraordinarily expensive, costing millions of dollars, and were owned exclusively by large governments, universities, and corporations.

Mainframes processed work in batches — operators loaded decks of punched cards containing programs and data, the machine executed the entire batch sequentially, and results were printed hours later. There was no interactive access. The machine worked on one job at a time, and users waited in queue for their turn.

TIME-SHARING — THE FIRST STEP TOWARD CLOUD:

The fundamental problem with batch processing was low utilisation and long turnaround times. In the early 1960s, computer scientists developed time-sharing operating systems that allowed multiple users to interact with the same mainframe simultaneously. The OS rapidly switches between user sessions (in milliseconds), giving each user the illusion of having exclusive access to the machine.

- **CTSS (Compatible Time-Sharing System, 1961):** Developed at MIT, CTSS was the first operational time-sharing system. It allowed 30 simultaneous users to interact with an IBM 7094 mainframe through terminals.
- **MULTICS (Multiplexed Information and Computing Service, 1969):** A highly influential time-sharing OS jointly developed by MIT, Bell Labs, and General Electric. MULTICS introduced many concepts still used today — hierarchical file systems, per-process security rings, and dynamic linking. Though MULTICS itself was not widely deployed, its ideas directly influenced the design of Unix.
- **Unix (1969):** Ken Thompson and Dennis Ritchie at Bell Labs created Unix as a simpler alternative to MULTICS. Unix's design philosophy — small tools, plain-text interfaces, and modularity — became the foundation for modern server operating systems including Linux, which underpins virtually all cloud infrastructure today.

John McCarthy — The Father of the Cloud Concept

- In 1961, at the centennial celebration of MIT, computer scientist John McCarthy (who also coined the term 'artificial intelligence') gave a speech in which he proposed that 'computation may someday be organised as a public utility, just as the telephone system is a public utility.' This visionary idea — computing delivered as a metered service to anyone who needs it — is the conceptual foundation of cloud computing, articulated more than 45 years before Amazon Web Services launched EC2. McCarthy's 1961 proposal is therefore considered the intellectual origin of both utility computing and cloud computing.

ERA 2 — PERSONAL COMPUTERS AND CLIENT-SERVER COMPUTING (1970S–1990S):

The Personal Computer Revolution:

The invention of the microprocessor — Intel's 4004 in 1971 and the 8080 in 1974 — made it possible to build computers small enough and inexpensive enough to be owned by individuals and small businesses. The Apple II (1977), the IBM PC (1981), and the IBM PC clones that followed rapidly democratised computing: millions of people now had their own machines.

The personal computer (PC) era represented a dramatic decentralisation of computing. Instead of sharing a powerful central mainframe, each person had their own processor, memory, and storage. This solved the problem of access — anyone could own a computer — but it created a new problem: data and applications became siloed on individual machines, and there was no easy way to share resources or collaborate.

Client-Server Computing:

The development of Local Area Networks (LANs) in the late 1970s and 1980s — notably Ethernet, standardised by Xerox, Intel, and DEC in 1980 — allowed PCs to be connected and share resources. This gave rise to the client-server architecture: a powerful centralised server provides shared resources (files, databases, print services, email), and client machines request and consume those services over the network.

Component	Role	Examples
File Server	Centralised storage shared across multiple clients	Novell NetWare, Windows Server File Services
Database Server	Runs DBMS; clients send SQL queries and receive results	Oracle Database, Microsoft SQL Server, MySQL
Print Server	Manages shared printer queue for multiple clients	Windows Print Server, CUPS (Linux)
Email Server	Stores, routes, and delivers email for all users	Microsoft Exchange, Sendmail, Postfix
Web Server	Serves web pages and application content to browsers	Apache HTTP Server, IIS, Nginx
Application Server	Hosts business logic for multi-tier enterprise applications	Oracle WebLogic, IBM WebSphere, JBoss

ERA 3 — THE INTERNET AND WEB ERA (1990S):

ARPANET TO THE INTERNET:

ARPANET (Advanced Research Projects Agency Network), funded by the US Department of Defense, went live in 1969 with four connected nodes — UCLA, Stanford Research Institute, UC Santa Barbara, and the University of Utah. It was the first wide-area packet-switched network, demonstrating that computers in different geographic locations could reliably exchange data.

Through the 1970s and 1980s, ARPANET grew and evolved. The adoption of TCP/IP as the standard networking protocol suite in 1983 (known as 'Flag Day') unified a collection of disparate networks into a single interconnected network — the Internet. By 1990, ARPANET was decommissioned; the civilian Internet it had spawned continued to grow exponentially.

THE WORLD WIDE WEB:

In 1989–1991, Tim Berners-Lee at CERN (the European particle physics laboratory) invented the World Wide Web — a system of hyperlinked documents accessible over the Internet using a standard protocol (HTTP) and a standard document format (HTML). The first web browser (WorldWideWeb, later Nexus) was released in 1991.

The Web transformed the Internet from a platform used primarily by academics and government researchers into a global information infrastructure accessible to ordinary people. The introduction of the Mosaic graphical browser in 1993 and Netscape Navigator in 1994 triggered explosive growth. By the mid-1990s, millions of businesses were establishing an online presence, and e-commerce was emerging as a viable business model.

ERA 4 — GRID AND CLUSTER COMPUTING (LATE 1990S–EARLY 2000S):

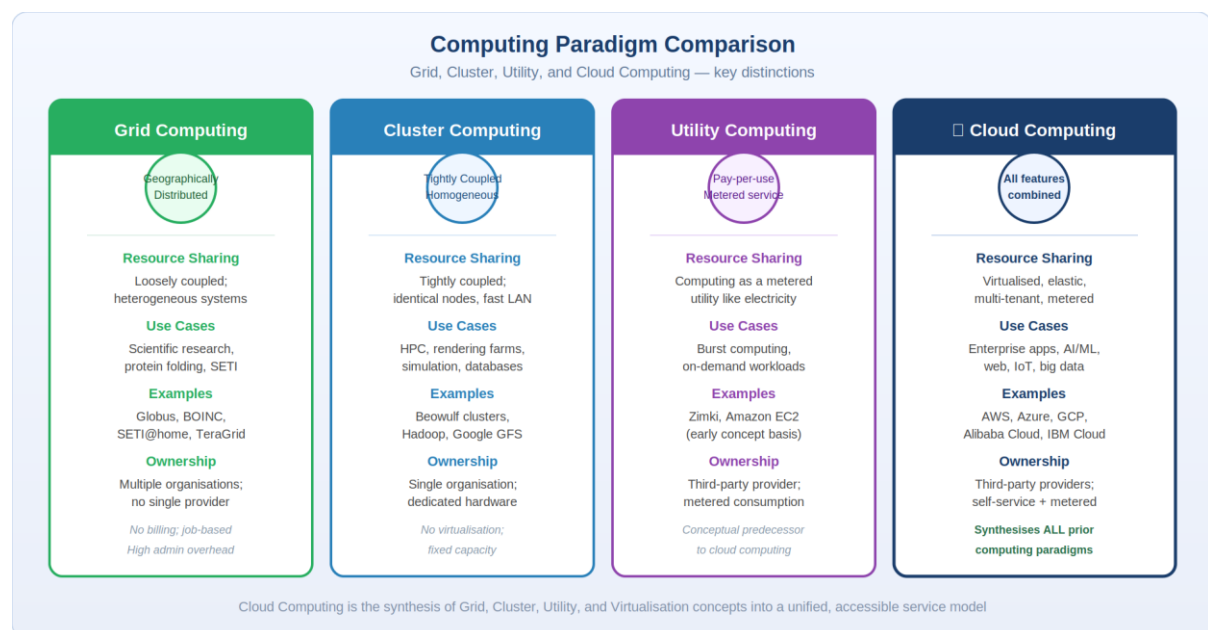


Figure : Computing Paradigm Comparison — Grid, Cluster, Utility, and Cloud Computing

Grid Computing

Grid computing emerged in the late 1990s as a way to harness the combined processing power of geographically distributed computers — across different organisations, cities, or even countries — to tackle large-scale computational problems that no single computer or cluster could solve efficiently. The term 'grid' was coined by analogy with the electrical power grid: just as electricity is available from a universal infrastructure wherever you plug in, the computing grid would make compute power available wherever and whenever it was needed.

Key Characteristics of Grid Computing

- **Heterogeneity:** A grid connects computers from different manufacturers, running different operating systems, owned by different organisations. Middleware (notably the Globus Toolkit) provides a common interface.
- **Geographic distribution:** Grid nodes are spread across cities, countries, or continents — connected via wide-area networks rather than high-speed LANs.
- **Resource sharing:** Organisations contribute their idle computing capacity to the grid and in return gain access to the combined capacity of all participants.
- **No centralised control:** Grids are typically governed by virtual organisations — consortia of institutions that agree to share resources under a common policy framework.
- **Job-based scheduling:** Workloads are submitted as discrete jobs, which the grid scheduler dispatches to available nodes. There is no persistent service concept — jobs run to completion and resources are released.

CLUSTER COMPUTING:

While grid computing focuses on loosely coupled, geographically distributed heterogeneous systems, cluster computing connects a set of tightly coupled, homogeneous computers within a single location (typically a single rack or room) using a high-speed local network. Clusters are designed for parallel processing of computationally intensive tasks with very low inter-node latency.

BEOWULF CLUSTERS:

In 1994, NASA scientists Thomas Sterling and Donald Becker built the first Beowulf cluster — a high-performance parallel computer assembled entirely from commodity off-the-shelf PC hardware (Intel 486 CPUs) connected by Ethernet. This proved that supercomputing performance was achievable at a fraction of the cost of proprietary supercomputers. The Beowulf design became highly influential and is the template for modern HPC (High-Performance Computing) clusters.

GOOGLE'S CLUSTER ARCHITECTURE:

In their landmark 2003 paper, Jeff Dean and Sanjay Ghemawat described Google's approach to large-scale distributed computing: thousands of commodity Linux servers, with reliability achieved entirely through software (automatic replication, job rescheduling, and failure detection) rather than expensive hardware redundancy. This 'design for failure' philosophy was directly adopted by Amazon in building AWS and is a defining principle of modern cloud infrastructure.

LIMITATIONS OF GRID AND CLUSTER COMPUTING:

Limitation	Grid Computing	Cluster Computing
Ease of use	Complex setup; requires specialist grid middleware	Easier within one org; still requires HPC expertise
Billing / accounting	No metered billing; resource time donated or allocated	No commercial billing model
Self-service	No self-service portal; job submission via CLI tools	No self-service; sysadmin provisions resources

Elasticity	Fixed contributing nodes; no dynamic scale-out	Fixed cluster size; hardware procurement to expand
Multi-tenancy	Organisational boundary sharing only	Typically single-tenant
Commercial model	Academic/research sharing; no pay-per-use	Internal cost allocation; no external billing

ERA 5 — UTILITY COMPUTING AND VIRTUALISATION (MID 2000S):

The Utility Computing Concept

Utility computing is the business model in which computing resources — processing, storage, network — are packaged as a metered service and sold to customers based on their actual consumption, exactly as utilities such as electricity, water, and telecommunications are sold. The customer pays only for what they use; the provider maintains the infrastructure and guarantees availability.

While John McCarthy proposed this idea in 1961, the technical infrastructure to implement it commercially did not exist until the mid-2000s. The combination of three enabling technologies made utility computing finally feasible at scale:

- **Broadband Internet:** High-speed, always-on Internet connectivity became widely available in the early 2000s, providing the network infrastructure to deliver computing as a remote service.
- **Virtualisation:** Mature server virtualisation software (VMware ESX Server launched in 2001; Xen open-sourced in 2003) allowed a single physical server to host multiple isolated virtual machines, enabling the multi-tenancy required for cost-effective utility computing.
- **Commodity hardware at scale:** The falling cost of x86 server hardware made it economically viable to build large pools of shared compute capacity and offer it at prices low enough for individual developers and small businesses.

ERA 6 — THE BIRTH OF MODERN CLOUD COMPUTING (2006–2012):

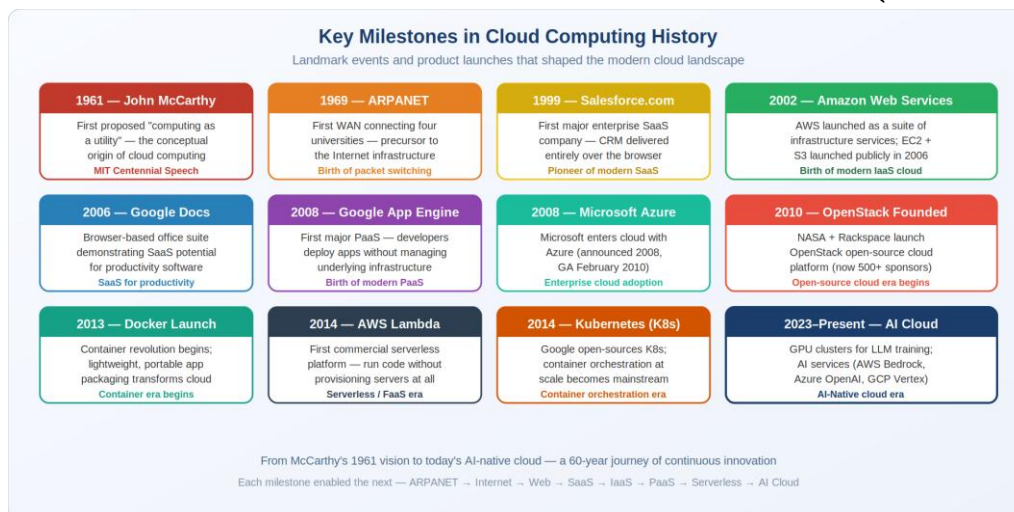


Figure : Key Milestones in Cloud Computing History — landmark product launches and events

AMAZON WEB SERVICES (2002–2006)

Amazon's journey to becoming the world's leading cloud provider began with an internal infrastructure crisis. As Amazon's e-commerce platform grew rapidly in the late 1990s and early 2000s, the engineering team struggled with the complexity of building new features on top of a tangled monolithic codebase and non-standardised internal infrastructure. In 2002, Amazon began a major internal restructuring — teams were required to expose their functionality through well-defined service APIs, and all inter-team communication had to go through those APIs (Jeff Bezos's famous 'API mandate').

This API-first restructuring transformed Amazon's internal infrastructure into a collection of discrete, independently scalable services. By 2003, Amazon's technology leaders recognised that this service-oriented infrastructure could be offered externally to other developers and businesses. Amazon Web Services launched publicly in 2002 as a set of free developer tools, but the landmark moment was the commercial launch of:

- **Amazon S3 (Simple Storage Service) — March 2006:** The first commercially successful cloud storage service. S3 offered scalable object storage at \$0.15 per GB per month with no upfront commitment — developers could store unlimited data and pay only for what they used.
- **Amazon EC2 (Elastic Compute Cloud) — Beta August 2006, GA October 2008:** Virtual machines in the cloud, available in minutes, billed per hour. EC2 allowed any developer or startup to rent a Linux server for \$0.10 per hour, eliminating the need to buy physical hardware.
- **Amazon SQS (Simple Queue Service) — 2004/2006:** A managed message queue service — the first AWS service, originally launched internally in 2004 and publicly in 2006.

GOOGLE AND THE PAAS MODEL (2006–2008)

Google's contribution to cloud computing came primarily through two channels: influential research papers and the launch of the first major Platform-as-a-Service offering.

- **Google File System (GFS, 2003):** Google published a paper describing GFS, a distributed file system designed to run on commodity hardware and tolerate individual node failures gracefully. GFS directly inspired Hadoop HDFS, which became the foundation of the big data ecosystem.
- **MapReduce (2004):** Google published the MapReduce programming model for processing large datasets in parallel across thousands of nodes. This paper inspired the Apache Hadoop project and the entire distributed data processing ecosystem.
- **Bigtable (2006):** Google's distributed NoSQL database paper inspired Apache HBase and influenced the design of many cloud databases.
- **Google App Engine (April 2008):** The first major commercial PaaS offering. App Engine allowed developers to deploy Python (later Java, Go, PHP) web applications without managing any servers. The platform automatically scaled applications, handled load balancing, and provided managed databases. It was truly serverless before the term existed.

- **Google Docs / Google Apps (2006):** Google's browser-based productivity suite demonstrated that enterprise-grade SaaS applications were viable at scale, competing directly with Microsoft Office.

MICROSOFT AZURE (2008–2010)

Microsoft announced the Azure cloud platform at its Professional Developers Conference in October 2008, achieving general availability in February 2010. Azure was initially a PaaS offering — it allowed .NET developers to build and deploy applications on Microsoft-managed infrastructure. Over time, Azure evolved into a full IaaS+PaaS hybrid, and today it is the second-largest cloud platform behind AWS.

Microsoft's entry was significant because it brought the world's largest enterprise software company into the cloud market, triggering widespread enterprise adoption. Azure's deep integration with Active Directory, SQL Server, Visual Studio, and Office 365 made it the default choice for enterprises already invested in the Microsoft ecosystem.

NIST DEFINITION AND STANDARDISATION (2011)

As cloud computing grew rapidly, the lack of a clear, authoritative definition created confusion — vendors labelled almost any Internet-delivered service as 'cloud.' In September 2011, NIST published Special Publication 800-145, providing the canonical definition of cloud computing with its five essential characteristics, three service models, and four deployment models. This definition became the industry standard and is still the authoritative reference used globally by researchers, practitioners, and regulators.

OPENSTACK (2010)

In July 2010, NASA and Rackspace jointly released OpenStack — an open-source cloud computing platform designed to provide all the capabilities of a commercial cloud (compute, storage, networking, identity, orchestration) as free, open-source software that any organisation could deploy on their own hardware. OpenStack provided a vendor-neutral alternative to proprietary cloud platforms and became the foundation for private and community clouds at telecommunications companies, governments, research institutions, and enterprises worldwide.

THE EVOLUTIONARY PATH — FROM TIME-SHARING TO CLOUD:

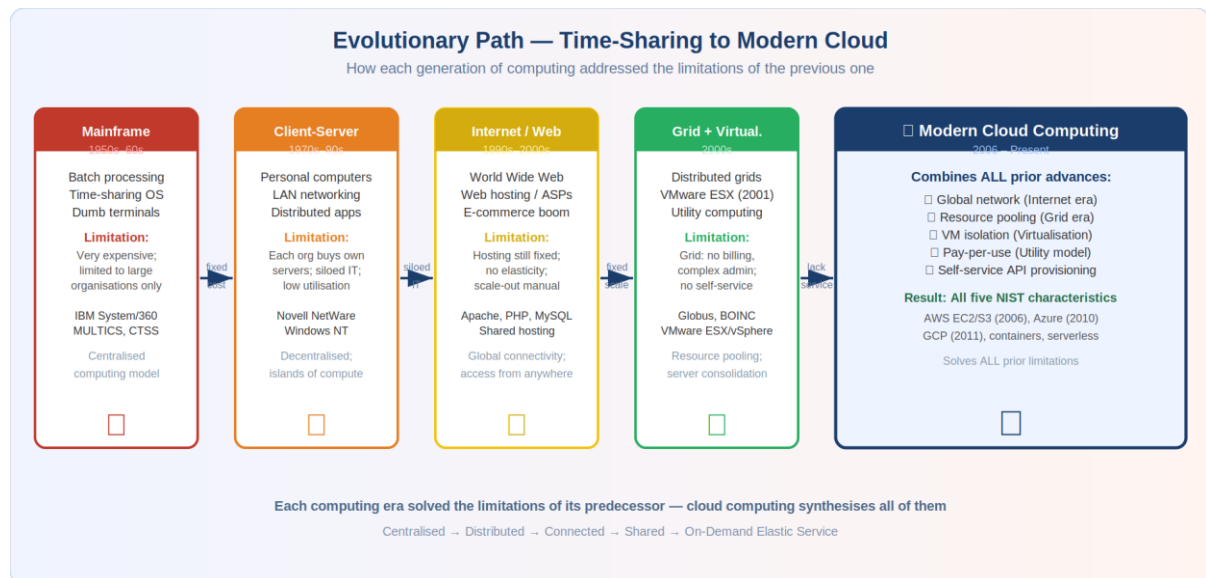


Figure : Computing Evolution Path — how each era addressed the limitations of its predecessor

Figure 4 synthesises the evolutionary journey into five stages, showing explicitly how each era solved the problems of the previous one and what problems it left for the next generation to address. Cloud computing sits at the end of this journey because it uniquely addresses the limitations of all prior paradigms simultaneously.

Era	Key Contribution to Cloud	Limitation Passed to Next Era
Mainframe / Time-Sharing (1950s–60s)	Concept: multiple users sharing one resource; time-sharing OS; remote terminal access	Extremely expensive; accessible only to large organisations; not interactive enough
Personal Computer / Client-Server (1970s–90s)	Distributed computing; LAN networking; TCP/IP protocols; client-server architecture; web servers	Siloed IT; low utilisation; no geographic reach; no elasticity; high CapEx per organisation
Internet / Web (1990s)	Global network connectivity; HTTP/HTTPS; browser-based access; ASP model; e-commerce scale challenges	Hosting still fixed; no elasticity; single-tenant; complex scaling; no pay-per-use
Grid / Cluster Computing (Late 1990s–2000s)	Resource pooling across organisations; commodity hardware at scale; design-for-failure philosophy; MapReduce/GFS	No billing; no self-service; complex admin; heterogeneity problems; no SLAs

Utility Computing / Virtualisation (Mid 2000s)	VM isolation; multi-tenancy; pay-per-use billing concept; server consolidation; API-driven provisioning	Early offerings too expensive; no developer self-service; no elastic scaling
Cloud Computing (2006–present)	Synthesises ALL prior eras: network access + resource pooling + virtualisation + utility billing + self-service + elasticity	Ongoing: sustainability; edge latency; regulatory/sovereignty concerns; vendor lock-in